Chapter 1. The Algebra and Topology of $\mathbb{R}^n$

Linear algebra, matrix inversion and determinants, metric topology, compactness and connectedness in $\mathbb{R}^n$

Contents

1 Review of Linear Algebra **2**
1.1 Vector spaces . . . 2
1.2 Inner products and norms . . . 3
1.3 Matrices . . . 4
1.4 Linear transformations . . . 4
1.5 Rank of a matrix . . . 5
1.6 Transpose . . . 5

2 Matrix Inversion and Determinants **6**
2.1 Elementary matrices . . . 6
2.2 The inverse of a matrix . . . 6
2.3 Determinants . . . 7
2.4 A formula for A^{-1} . . . 9
2.5 Expansion by cofactors . . . 11

3 Review of Topology in $\mathbb{R}^n$ **11**
3.1 Metric spaces . . . 11
3.2 Limits and continuity . . . 13
3.3 Interior and exterior . . . 14

4 Compact Subspaces and Connected Subspaces of $\mathbb{R}^n$ **14**
4.1 Compact spaces . . . 14
4.2 Connected spaces . . . 17

Overview

This chapter collects the linear algebra and the topology of $\mathbb{R}^n$ used throughout. Sections 1 and 3 are review; Sections 2 and 4 contain material that may be new (elementary matrices, the axiomatic determinant, the ϵ-neighborhood theorem, compactness of rectangles). Results that Munkres states without proof are given short proofs here, marked [Supplement]. Elements of $\mathbb{R}^n$ are written as column matrices unless stated otherwise.

§1. Review of Linear Algebra

Motivation. Multivariable calculus is linear algebra applied locally: a derivative is a linear map, a change of variables multiplies volume by a determinant, and differential forms are alternating multilinear maps. We therefore begin by fixing the linear-algebra facts the rest of the book uses.

1.1 Vector spaces

Definition 1.1 (vector space)

A set V with an operation $\mathbf{x}+\mathbf{y}$ (vector addition) and an operation $c\mathbf{x}$ (scalar multiplication by $c \in \mathbb{R}$) is a **vector space** (or **linear space**) if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $c, d \in \mathbb{R}$:

(1) $\mathbf{x}+\mathbf{y} = \mathbf{y}+\mathbf{x}$;
(2) $\mathbf{x}+(\mathbf{y}+\mathbf{z}) = (\mathbf{x}+\mathbf{y})+\mathbf{z}$;
(3) there is a unique vector $\mathbf{0}$ with $\mathbf{x}+\mathbf{0} = \mathbf{x}$ for all $\mathbf{x}$;
(4) $\mathbf{x}+(-1)\mathbf{x} = \mathbf{0}$;
(5) $1\mathbf{x} = \mathbf{x}$;
(6) $c(d\mathbf{x}) = (cd)\mathbf{x}$;
(7) $(c+d)\mathbf{x} = c\mathbf{x}+d\mathbf{x}$;
(8) $c(\mathbf{x}+\mathbf{y}) = c\mathbf{x}+c\mathbf{y}$.

The basic example is $\mathbb{R}^n$, the set of n-tuples $\mathbf{x} = (x_1, \dots, x_n)$ with componentwise operations $\mathbf{x}+\mathbf{y} = (x_1+y_1, \dots, x_n+y_n)$, $c\mathbf{x} = (cx_1, \dots, cx_n)$.

A subset $W \subset V$ is a **linear subspace** if $\mathbf{x}+\mathbf{y} \in W$ and $c\mathbf{x} \in W$ whenever $\mathbf{x}, \mathbf{y} \in W$, $c \in \mathbb{R}$; then W is a vector space in its own right. In the first part of the book only $\mathbb{R}^n$ and its subspaces occur.

Definition 1.2 (span, independence, basis, dimension)

Let $\mathbf{a}_1, \dots, \mathbf{a}_m \in V$.

- They **span** V if every $\mathbf{x} \in V$ can be written in *at least* one way as a **linear combination** $\mathbf{x} = c_1\mathbf{a}_1 + \cdots + c_m\mathbf{a}_m$.
- They are **independent** if every $\mathbf{x} \in V$ can be written in *at most* one way as such a combination; equivalently, $d_1\mathbf{a}_1 + \cdots + d_m\mathbf{a}_m = \mathbf{0}$ forces $d_1 = \cdots = d_m = 0$.
- They form a **basis** if they both span V and are independent.

If V has a basis of m vectors, m is the **dimension** of V (by Theorem 1.1 this is well defined); the zero space has dimension 0.

Theorem 1.1

Suppose V has a basis consisting of m vectors. Then any set of vectors that spans V has at least m vectors, and any set of vectors of V that is independent has at most m vectors. In particular, any basis for V has exactly m vectors.

Proof [Supplement]. It suffices to show: if $\mathbf{w}_1, \dots, \mathbf{w}_k$ span V and $\mathbf{v}_1, \dots, \mathbf{v}_l$ are independent, then $l \le k$ (apply it with the basis as the spanning set, and with the basis as the independent set). Write $\mathbf{v}_j = \sum_{i=1}^{k} a_{ij}\mathbf{w}_i$. If $l > k$, the homogeneous linear system $\sum_{j=1}^{l} a_{ij}x_j = 0$ $(i = 1, \dots, k)$ has more unknowns than equations, hence (by Gauss–Jordan reduction: an echelon form has at most k corner entries, so some unknown is free) a nonzero solution $(x_1, \dots, x_l)$. Then $\sum_j x_j\mathbf{v}_j = \sum_i(\sum_j a_{ij}x_j)\mathbf{w}_i = \mathbf{0}$, contradicting independence. ■

$\mathbb{R}^n$ has dimension n: the **standard basis** is $\mathbf{e}_1 = (1, 0, \dots, 0)$, $\mathbf{e}_2 = (0, 1, 0, \dots, 0)$, $\dots$, $\mathbf{e}_n = (0, \dots, 0, 1)$. Every basis of $\mathbb{R}^n$ has exactly n vectors. (Spanning, independence and bases make sense for infinite sets of vectors too, but we shall not need this.)

Theorem 1.2

Let V be a vector space of dimension m. If W is a linear subspace of V different from V, then W has dimension less than m. Furthermore, any basis $\mathbf{a}_1, \dots, \mathbf{a}_k$ for W may be extended to a basis $\mathbf{a}_1, \dots, \mathbf{a}_k, \mathbf{a}_{k+1}, \dots, \mathbf{a}_m$ for V.

Proof [Supplement]. First, W has a finite basis: choose independent vectors of W one at a time as long as possible; by Theorem 1.1 the process stops after at most m steps, and a maximal independent subset of W spans W (if $\mathbf{x} \in W$ were not a combination of them, adjoining $\mathbf{x}$ would keep independence). Now let $\mathbf{a}_1, \dots, \mathbf{a}_k$ be a basis of W. If they do not span V, pick $\mathbf{a}_{k+1} \in V$ not in their span; then $\mathbf{a}_1, \dots, \mathbf{a}_{k+1}$ are independent (a relation with nonzero coefficient of $\mathbf{a}_{k+1}$ would put $\mathbf{a}_{k+1}$ in the span; one with zero coefficient contradicts independence of the $\mathbf{a}_i$). Repeat; by Theorem 1.1 the process stops when we have an independent spanning set of m vectors. If $W \neq V$, at least one vector was added, so $k < m$. ■

1.2 Inner products and norms

Definition 1.3 (inner product)

An **inner product** on V assigns to $\mathbf{x}, \mathbf{y} \in V$ a real number $\langle \mathbf{x}, \mathbf{y} \rangle$ such that for all $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and scalars c:
(1) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$;
(2) $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$;
(3) $\langle c\mathbf{x}, \mathbf{y} \rangle = c\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, c\mathbf{y} \rangle$;
(4) $\langle \mathbf{x}, \mathbf{x} \rangle > 0$ if $\mathbf{x} \neq \mathbf{0}$. V together with an inner product is an **inner product space**.

On $\mathbb{R}^n$ we use the **dot product** $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + \cdots + x_ny_n$ (written $\langle \mathbf{x}, \mathbf{y} \rangle$ rather than $\mathbf{x} \cdot \mathbf{y}$ to avoid confusion with the matrix product).

Definition 1.4 (norm)

The **length** (or norm) in an inner product space is $\|\mathbf{x}\| = \langle \mathbf{x}, \mathbf{x} \rangle^{1/2}$. It satisfies (1) $\|\mathbf{x}\| > 0$ if $\mathbf{x} \neq \mathbf{0}$;
(2) $\|c\mathbf{x}\| = |c|\,\|\mathbf{x}\|$;
(3) $\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ (**triangle inequality**); equivalently (3′) $\|\mathbf{x} - \mathbf{y}\| \ge \|\mathbf{x}\| - \|\mathbf{y}\|$. Any function $V \to \mathbb{R}$ with properties (1)–(3) is called a **norm** on V.

Proof of (3) [Supplement]. *Cauchy–Schwarz*: $|\langle \mathbf{x},\mathbf{y}\rangle| \le \|\mathbf{x}\|\|\mathbf{y}\|$. This is trivial if $\mathbf{x}$ or $\mathbf{y}$ is $\mathbf{0}$; otherwise put $c = 1/\|\mathbf{x}\|$, $d = 1/\|\mathbf{y}\|$; then $0 \le \|c\mathbf{x} \pm d\mathbf{y}\|^2 = 2 \pm 2cd\langle \mathbf{x},\mathbf{y}\rangle$, so $|\langle \mathbf{x},\mathbf{y}\rangle| \le 1/(cd)$. Hence $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x},\mathbf{y}\rangle + \|\mathbf{y}\|^2 \le (\|\mathbf{x}\| + \|\mathbf{y}\|)^2$. Applying (3) to $\mathbf{x} = (\mathbf{x}-\mathbf{y}) + \mathbf{y}$ gives (3′). ∎

On $\mathbb{R}^n$ we have the **euclidean norm** $\|\mathbf{x}\|$ coming from the dot product, and the **sup norm**

$$|\mathbf{x}| = \max\{|x_1|, \dots, |x_n|\}.$$

The sup norm is often more convenient. The two are related by

$$|\mathbf{x}| \le \|\mathbf{x}\| \le \sqrt{n}\,|\mathbf{x}|, \tag{1.1}$$

since $\max_i x_i^2 \le \sum_i x_i^2 \le n \max_i x_i^2$.

1.3 Matrices

An n by m matrix A has entries a_{ij} (i = row index, j = column index). With entrywise addition and scalar multiplication the n by m matrices form a vector space (essentially $\mathbb{R}^{nm}$). If A is n by m and B is m by p, the **product** $C = A \cdot B$ is the n by p matrix with

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

It satisfies $A \cdot (B \cdot C) = (A \cdot B) \cdot C$, $A \cdot (B + C) = A \cdot B + A \cdot C$, $(A + B) \cdot C = A \cdot C + B \cdot C$, $(cA) \cdot B = c(A \cdot B) = A \cdot (cB)$, and $I_n \cdot A = A = A \cdot I_m$ for A of size n by m, where the **identity matrix** I_k has entries δ_{ij} ($\delta_{ij} = 1$ if $i = j$, 0 otherwise).

The sup norm extends to matrices: $|A| = \max\{|a_{ij}| : i = 1, \dots, n,\ j = 1, \dots, m\}$; it is a norm.

Theorem 1.3

If A has size n by m and B has size m by p, then $|A \cdot B| \le m|A|\,|B|$.

Proof [Supplement]. $|c_{ij}| = |\sum_{k=1}^m a_{ik}b_{kj}| \le \sum_{k=1}^m |A||B| = m|A||B|$ for every i, j. ∎

1.4 Linear transformations

Definition 1.5 (linear transformation)

A function $T : V \to W$ between vector spaces is a **linear transformation** if $T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ and $T(c\mathbf{x}) = cT(\mathbf{x})$. If moreover T is a bijection, T is a **linear isomorphism**.

Composites of linear transformations are linear, and the inverse of a linear isomorphism is linear.

Theorem 1.4

Let V be a vector space with basis $\mathbf{a}_1, \dots, \mathbf{a}_m$. Let W be a vector space. Given any m vectors $\mathbf{b}_1, \dots, \mathbf{b}_m$ in W, there is exactly one linear transformation $T : V \to W$ such that $T(\mathbf{a}_i) = \mathbf{b}_i$ for all i.

Proof [Supplement]. Each $\mathbf{x} \in V$ has a unique expression $\mathbf{x} = \sum c_i \mathbf{a}_i$; define $T(\mathbf{x}) = \sum c_i \mathbf{b}_i$. Uniqueness of the coefficients makes T well defined, and since the coefficients of $\mathbf{x}+\mathbf{y}$ and $c\mathbf{x}$ are the sums and multiples of those of $\mathbf{x}, \mathbf{y}$, T is linear. Any linear T' with $T'(\mathbf{a}_i) = \mathbf{b}_i$ satisfies $T'(\sum c_i \mathbf{a}_i) = \sum c_i \mathbf{b}_i$, so $T' = T$. ∎

The maps $(x_1, \dots, x_n) \mapsto [x_1 \ \cdots \ x_n]$ (**row matrix**) and $(x_1, \dots, x_n) \mapsto$ the n by 1 **column matrix** with entries x_i are linear isomorphisms. Representing elements of $\mathbb{R}^m$ as column matrices, a fixed n by m matrix A defines a linear map $T : \mathbb{R}^m \to \mathbb{R}^n$, $T(\mathbf{x}) = A \cdot \mathbf{x}$. Conversely every linear $T : \mathbb{R}^m \to \mathbb{R}^n$ has this form: let $\mathbf{b}_j = T(\mathbf{e}_j)$ and $A = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_m]$; then $A \cdot \mathbf{e}_j = \mathbf{b}_j = T(\mathbf{e}_j)$ for all j, so $A \cdot \mathbf{x} = T(\mathbf{x})$ by Theorem 1.4.

Convention

Throughout, the elements of $\mathbb{R}^n$ are represented by column matrices, unless specifically stated otherwise.

1.5 Rank of a matrix

For an n by m matrix A, the **column space** is the subspace of $\mathbb{R}^n$ spanned by the columns; its dimension is the **column rank** ($\leq \min(m, n)$). The **row space** is the subspace of $\mathbb{R}^m$ spanned by the rows; its dimension is the **row rank**.

Theorem 1.5

For any matrix A, the row rank of A equals the column rank of A.

Proof [Supplement]. Let r be the column rank and let $\mathbf{c}_1, \dots, \mathbf{c}_r \in \mathbb{R}^n$ be a basis of the column space; let $C = [\mathbf{c}_1 \ \cdots \ \mathbf{c}_r]$ (n by r). Each column of A is a combination of the $\mathbf{c}_k$, so $A = C \cdot D$ for some r by m matrix D. Then each row of A is a combination of the r rows of D (row i of A is $\sum_k c_{ik}\cdot$(row k of D)), so the row rank of A is $\leq r$. Applying this to A^{tr} (whose columns are the rows of A) gives the reverse inequality. ■

The common value is the **rank** of A.

Elementary row operations on a matrix are:

(1) exchange rows $i_1 \neq i_2$;

(2) replace row i_1 by itself plus c times row i_2 ($i_1 \neq i_2$);

(3) multiply row i by a scalar $\lambda \neq 0$. Each is invertible, and its inverse is an elementary operation of the same type.

Theorem 1.6

If B is the matrix obtained by applying an elementary row operation to A, then $\operatorname{rank} B = \operatorname{rank} A$.

Proof [Supplement]. Each row of B is a linear combination of rows of A, so the row space of B is contained in that of A. Since A is obtained from B by the inverse elementary operation, the reverse inclusion holds too. The row spaces coincide, so the row ranks agree. ■

Gauss–Jordan reduction applies elementary operations to bring A to **echelon form** (stairstep form): below the "stairsteps" all entries are 0 and the corner entries (pivots) are nonzero. Operations of types (1) and (2) suffice. The nonzero rows of a matrix in echelon form are independent, so its rank is the number of nonzero rows. Using further operations of types (2) and (3) one reaches the **reduced echelon form**, in which each corner entry is 1 and all entries directly above it are 0.

1.6 Transpose

The **transpose** of the n by m matrix A is the m by n matrix A^{tr} with entries $d_{ij} = a_{ji}$. Then $(A^{\mathrm{tr}})^{\mathrm{tr}} = A$, $(A + B)^{\mathrm{tr}} = A^{\mathrm{tr}} + B^{\mathrm{tr}}$, $(A \cdot C)^{\mathrm{tr}} = C^{\mathrm{tr}} \cdot A^{\mathrm{tr}}$, and $\operatorname{rank} A^{\mathrm{tr}} = \operatorname{rank} A$ (the row rank of

A^{tr} is the column rank of A).

§2. Matrix Inversion and Determinants

Motivation. The determinant will appear as the volume-scaling factor in the change of variables theorem and as the model for alternating tensors. Here we characterize it by its row properties, which gives the product formula, the inverse formula and cofactor expansion in one framework.

2.1 Elementary matrices

Definition 2.1 (elementary matrix)

An **elementary matrix** of size n by n is the matrix obtained by applying one of the elementary row operations to I_n.

There are three types: E (rows i_1, i_2 of I_n exchanged), E' (I_n with an extra entry c in row i_1, column i_2), and E'' (I_n with the (i,i) entry replaced by $\lambda \neq 0$).

Theorem 2.1

Let A be an n by m matrix. Any elementary row operation on A may be carried out by premultiplying A by the corresponding elementary matrix.

Proof. Direct computation: multiplying A on the left by E interchanges rows i_1 and i_2 of A; by E' replaces row i_1 by itself plus c times row i_2; by E'' multiplies row i by λ. ■

2.2 The inverse of a matrix

Definition 2.2 (left and right inverse)

Let A be n by m, and B, C be m by n. B is a **left inverse** for A if $B \cdot A = I_m$; C is a **right inverse** for A if $A \cdot C = I_n$.

Theorem 2.2

If A has both a left inverse B and a right inverse C, then they are unique and equal.

Proof. $C = I_m \cdot C = (B \cdot A) \cdot C = B \cdot (A \cdot C) = B \cdot I_n = B$. If B_1 is another left inverse, the same computation gives $C = B_1$, so $B_1 = B$; similarly C is unique. ■

Definition 2.3 (invertible)

If A has both a left and a right inverse, A is **invertible**; the unique two-sided inverse is denoted A^{-1}.

Theorem 2.3

Let A be a matrix of size n by m. If A is invertible, then $n = m = \operatorname{rank} A$.

Proof.

Step 1. For any k by n matrix D, $\operatorname{rank}(D \cdot A) \leq \operatorname{rank} A$. Indeed, if R is a 1 by n row matrix, $R \cdot A$ is a linear combination of the rows of A. Each row of $D \cdot A$ is (row of D)$\cdot A$, so the row space of $D \cdot A$ lies in the row space of A.

Step 2. If A has a left inverse B, then $\operatorname{rank} A = m$ (the number of columns). Indeed $m = \operatorname{rank} I_m = \operatorname{rank}(B \cdot A) \leq \operatorname{rank} A$ by Step 1, while the row space of A is a subspace of $\mathbb{R}^m$, so $\operatorname{rank} A \leq m$.

Step 3. Let $B = A^{-1}$. Since B is a left inverse, $\operatorname{rank} A = m$ by Step 2. Since B is a right inverse, $B^{\mathrm{tr}} \cdot A^{\mathrm{tr}} = I_n^{\mathrm{tr}} = I_n$, so B^{tr} is a left inverse of the m by n matrix A^{tr} and Step 2 gives $\operatorname{rank} A = \operatorname{rank} A^{\mathrm{tr}} = n$. ■

Theorem 2.4

Let A be a matrix of size n by m with $n = m = \operatorname{rank} A$. Then A is invertible; furthermore, A equals a product of elementary matrices.

Proof.

Step 1. Every elementary matrix is invertible with elementary inverse: E is its own inverse, E' with c replaced by $-c$ inverts E', and E'' with λ replaced by $1/\lambda$ inverts E''.

Step 2. Reduce A to reduced echelon form C by elementary operations. C is square of rank n, so it has n nonzero rows; hence $C = I_n$. By Theorem 2.1 there are elementary matrices $E_1, \ldots, E_k$ with $E_k \cdots E_2 E_1 A = I_n$. Multiplying on the left successively by $E_k^{-1}, \ldots, E_1^{-1}$ gives

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

a product of elementary matrices, and $B = E_k E_{k-1} \cdots E_1$ is both a left and a right inverse for A. ■

Theorem 2.5

If A is a square matrix and B is a left inverse for A, then B is also a right inverse for A.

Proof. By Step 2 of the proof of Theorem 2.3, $\operatorname{rank} A$ equals the number of columns, which (as A is square) equals the number of rows. By Theorem 2.4, A has an inverse, which equals B by Theorem 2.2. ■

An n by n matrix is **singular** if $\operatorname{rank} A < n$, **non-singular** otherwise. By the above, A is invertible iff it is non-singular.

2.3 Determinants

Since $|A|$ denotes the sup norm, the determinant is written $\det A$. We characterize it by axioms; the existence of such a function for all n is proved in Chapter 6 (§27).

Definition 2.4 (determinant function)

A function assigning to each n by n matrix A a real number $\det A$ is a **determinant function** if:

(1) if B is obtained from A by exchanging two rows, then $\det B = -\det A$;

(2) for each i, $\det A$ is linear as a function of the ith row alone: if $A_i(\mathbf{x})$ denotes A with row i replaced by $\mathbf{x}$, then $\det A_i(a\mathbf{x} + b\mathbf{y}) = a \det A_i(\mathbf{x}) + b \det A_i(\mathbf{y})$;

(3) $\det I_n = 1$.

Example 2.1

For 1 by 1 matrices $\det[a] = a$; for 2 by 2 matrices $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$; for 3 by 3,

$$\det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1 - a_1b_3c_2 - a_2b_1c_3.$$

These satisfy the axioms. For larger matrices a direct formula has $n!$ terms (24 for $n = 4$, 120 for $n = 5$), so a less direct approach is needed.

Theorem 2.6

Let A be an n by n matrix.

(a) If E is the elementary matrix exchanging rows i_1 and i_2, then $\det(E \cdot A) = -\det A$.

(b) If E' replaces row i_1 of A by itself plus c times row i_2, then $\det(E' \cdot A) = \det A$.

(c) If E'' multiplies row i of A by $\lambda \neq 0$, then $\det(E'' \cdot A) = \lambda \det A$.

(d) If $A = I_n$, then $\det A = 1$.

Proof. (a) is Axiom 1 and (d) is Axiom 3;

(c) is the special case $\det A_i(\lambda \mathbf{x}) = \lambda \det A_i(\mathbf{x})$ of Axiom 2. For (b): if A has two equal rows, exchanging them does not change A but changes the sign of $\det A$, so $\det A = 0$. Let $i = i_1$, $\mathbf{x} =$ row i_1, $\mathbf{y} =$ row i_2. Then

$$\det(E' \cdot A) = \det A_i(\mathbf{x} + c\mathbf{y}) = \det A_i(\mathbf{x}) + c \det A_i(\mathbf{y}) = \det A_i(\mathbf{x}) = \det A,$$

since $A_i(\mathbf{y})$ has two equal rows. ■

Taking $A = I_n$: $\det E = -1$, $\det E' = 1$, $\det E'' = \lambda$.

Theorem 2.7

Let A be a square matrix. If the rows of A are independent, then $\det A \neq 0$; if the rows are dependent, then $\det A = 0$. Thus an n by n matrix A has rank n if and only if $\det A \neq 0$.

Proof. If row i of A is zero, then $\det A = 0$: multiplying that row by 2 leaves A unchanged but doubles $\det A$. An elementary operation multiplies the determinant by -1, 1 or $\lambda \neq 0$, so it does not affect whether $\det A$ vanishes. Reduce A to echelon form B (types (1), (2)). If the rows of A are dependent, $\operatorname{rank} B = \operatorname{rank} A < n$, so B has a zero row, $\det B = 0$, and $\det A = 0$. If they are independent, reduce further to reduced echelon form C; being square of rank n, $C = I_n$, so $\det C = 1 \neq 0$ and $\det A \neq 0$. ■

Theorem 2.8

Given a square matrix A, reduce it to echelon form B by elementary row operations of types (1) and (2). If B has a zero row, then $\det A = 0$. Otherwise, let k be the number of row exchanges involved; then $\det A = (-1)^k$ times the product of the diagonal entries of B.

Proof. If B has a zero row, $\operatorname{rank} A < n$ and $\det A = 0$. Otherwise $\det A = (-1)^k \det B$ by Theorem 2.6(a),(b), and B is upper triangular with nonzero diagonal entries $b_{11}, \ldots, b_{nn}$. Using type (2) operations, clear the entries above the diagonal; the diagonal is unaffected, giving a diagonal matrix

C with $\det C = \det B$. Multiplying row i of C by $1/b_{ii}$ for each i yields I_n, so by Theorem 2.6(c),(d), $1 = \det I_n = (1/b_{11}) \cdots (1/b_{nn}) \det C$. Hence $\det B = \det C = b_{11} \cdots b_{nn}$. ■

Corollary 2.9

The determinant function is uniquely characterized by its three axioms. It is also characterized by the four properties listed in Theorem 2.6.

Proof. The calculation of $\det A$ in Theorem 2.8 uses only properties (a)–(d) of Theorem 2.6, which follow from the axioms. ■

Theorem 2.10

Let A and B be n by n matrices. Then $\det(A \cdot B) = (\det A) \cdot (\det B)$.

Proof.

Step 1. The theorem holds when A is elementary: $\det(E \cdot B) = -\det B = (\det E)(\det B)$, $\det(E' \cdot B) = \det B = (\det E')(\det B)$, $\det(E'' \cdot B) = \lambda \det B = (\det E'')(\det B)$.

Step 2. If $\operatorname{rank} A = n$, $A = E_1 \cdots E_k$ is a product of elementary matrices (Theorem 2.4), and repeated use of Step 1 gives $\det(A \cdot B) = (\det E_1) \cdots (\det E_k)(\det B)$ for all B. With $B = I_n$ this reads $\det A = (\det E_1) \cdots (\det E_k)$, and the theorem follows.

Step 3. If $\operatorname{rank} A < n$, then $\operatorname{rank}(A \cdot B) = \operatorname{rank}(B^{\mathrm{tr}} \cdot A^{\mathrm{tr}}) \leq \operatorname{rank} A^{\mathrm{tr}} < n$ by Step 1 of Theorem 2.3, so both sides vanish by Theorem 2.7. ■

Theorem 2.11

$\det A^{\mathrm{tr}} = \det A$.

Proof.

Step 1. For elementary matrices: $E^{\mathrm{tr}} = E$, $(E'')^{\mathrm{tr}} = E''$, and $(E')^{\mathrm{tr}}$ is again of type (2), so both have determinant 1.

Step 2. If $\operatorname{rank} A = n$, write $A = E_1 \cdots E_k$. Then by Theorem 2.10 and Step 1,

$$\det A^{\mathrm{tr}} = \det(E_k^{\mathrm{tr}} \cdots E_1^{\mathrm{tr}}) = (\det E_k^{\mathrm{tr}}) \cdots (\det E_1^{\mathrm{tr}}) = (\det E_1) \cdots (\det E_k) = \det A.$$

Step 3. If $\operatorname{rank} A < n$, then $\operatorname{rank} A^{\mathrm{tr}} < n$ and both determinants are 0. ■

2.4 A formula for A^{-1}

Definition 2.5 (minor, cofactor)

For an n by n matrix A, the $(n-1)$ by $(n-1)$ matrix A_{ij} obtained by deleting row i and column j is the (i,j)**-minor**; the number $(-1)^{i+j} \det A_{ij}$ is the (i,j)**-cofactor** of A.

Lemma 2.12

Let A be an n by n matrix and let b denote its entry in row i and column j.

(a) If all the entries in row i other than b vanish, then $\det A = b(-1)^{i+j} \det A_{ij}$.

(b) The same holds if all the entries in column j other than b vanish.

Proof.

Step 1. Fix $b, a_2, \dots, a_n$. For an $(n-1)$ by $(n-1)$ matrix D let $A(D)$ be the n by n matrix with first row $(b, a_2, \dots, a_n)$, first column $(b, 0, \dots, 0)^{\mathrm{tr}}$ and lower right block D. We show $\det A(D) = b \det D$. If $b = 0$, $\operatorname{rank} A(D) < n$ and both sides vanish. If $b \neq 0$, put $f(D) = (1/b) \det A(D)$. Exchanging two rows of D, adding a multiple of one row of D to another, or multiplying a row of D by λ has the same effect on the corresponding rows $(i+1)$ of $A(D)$, so f changes by -1, 1, λ, respectively. If $D = I_{n-1}$, $A(D)$ is in echelon form and $\det A(D) = b$ by Theorem 2.8, so $f(I_{n-1}) = 1$. By Corollary 2.9, $f(D) = \det D$.

Step 2. Taking transposes (Theorem 2.11), the same formula holds for the matrix with first row $(b, 0, \dots, 0)$, first column $(b, a_2, \dots, a_n)^{\mathrm{tr}}$ and block D.

Step 3. In the general case, $i-1$ exchanges of adjacent rows bring row i to the top without changing the order of the other rows; then $j-1$ exchanges of adjacent columns bring column j to the left edge. The resulting matrix C is of the form in Step 1 or 2, and its $(1,1)$-minor C_{11} equals A_{ij}. Each row exchange, and (by Theorem 2.11) each column exchange, changes the sign of the determinant, so

$$\det A = (-1)^{(i-1)+(j-1)} \det C = (-1)^{i+j} b \det C_{11} = (-1)^{i+j} b \det A_{ij}.$$

■

Theorem 2.13 (Cramer's rule)

Let A be an n by n matrix with successive columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{c} = (c_1, \dots, c_n)$ be column matrices. If $A \cdot \mathbf{x} = \mathbf{c}$, then

$$(\det A) \cdot x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n].$$

Proof. Let $C = [\mathbf{e}_1 \cdots \mathbf{e}_{i-1}\ \mathbf{x}\ \mathbf{e}_{i+1} \cdots \mathbf{e}_n]$. Since $A \cdot \mathbf{e}_j = \mathbf{a}_j$ and $A \cdot \mathbf{x} = \mathbf{c}$, we have $A \cdot C = [\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$, so by Theorem 2.10, $(\det A)(\det C)$ equals the right-hand side. C is I_n except for column i, which is $\mathbf{x}$; expanding by Lemma 2.12 along row i (whose only nonzero entry may be x_i in column i) gives $\det C = x_i (-1)^{i+i} \det I_{n-1} = x_i$. ■

Theorem 2.14

Let A be an n by n matrix of rank n; let $B = A^{-1}$. Then

$$b_{ij} = \frac{(-1)^{j+i} \det A_{ji}}{\det A}.$$

Proof. Fix j and let $\mathbf{x}$ be the jth column of B. Then $A \cdot B = I_n$ gives $A \cdot \mathbf{x} = \mathbf{e}_j$, and Cramer's rule gives $(\det A) x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{e}_j\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$. Column i of this matrix has a single nonzero entry 1 in row j, so by Lemma 2.12(b) the determinant is $1 \cdot (-1)^{j+i} \det A_{ji}$. Since $x_i = b_{ij}$, the theorem follows. ■

This yields the algorithm:

(1) form the **matrix of cofactors**, with (i,j) entry $(-1)^{i+j} \det A_{ij}$;

(2) transpose it;

(3) divide each entry by $\det A$. Its importance is theoretical (for instance it shows that the entries of A^{-1} are continuous functions of those of A); for computation, Gauss–Jordan reduction is more efficient.

2.5 Expansion by cofactors

Theorem 2.15

Let A be an n by n matrix and fix i. Then

$$\det A = \sum_{k=1}^{n} (-1)^{i+k} a_{ik} \det A_{ik}.$$

Proof. Let $A_i(\mathbf{x})$ be A with row i replaced by the row matrix $\mathbf{x}$. Row i of A is $\sum_k a_{ik}\mathbf{e}_k$, so by linearity (Axiom 2) and Lemma 2.12,

$$\det A = \sum_{k=1}^{n} a_{ik} \det A_i(\mathbf{e}_k) = \sum_{k=1}^{n} a_{ik}(-1)^{i+k} \det A_{ik}.$$

■

This is the expansion of the determinant by cofactors of the ith row; taking transposes gives expansion by cofactors of the jth column.

§3. Review of Topology in $\mathbb{R}^n$

Motivation. Differentiability and integrability are defined through limits, so we need open sets, closed sets and continuity in $\mathbb{R}^n$; the general metric-space language costs nothing extra and is reused later for manifolds.

3.1 Metric spaces

Definition 3.1 (metric space)

A **metric** on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:
(1) $d(x, y) = d(y, x)$;
(2) $d(x, y) \geq 0$, with equality iff $x = y$;
(3) $d(x, z) \leq d(x, y) + d(y, z)$. A **metric space** is a set with a specific metric. A subset $Y \subset X$ with the restricted metric is a **subspace** of X.

On $\mathbb{R}^n$ we have the **euclidean metric** $\|\mathbf{x} - \mathbf{y}\|$ and the **sup metric** $|\mathbf{x} - \mathbf{y}|$. $\mathbb{R}^n$ is called n-dimensional **euclidean space**.

Definition 3.2 (open and closed sets)

For $x_0 \in X$ and $\epsilon > 0$, the **ϵ-neighborhood** of x_0 is $U(x_0; \epsilon) = \{x : d(x, x_0) < \epsilon\}$. A set $U \subset X$ is **open** in X if for each $x_0 \in U$ some $U(x_0; \epsilon) \subset U$; a set C is **closed** in X if $X - C$ is open. Any open set containing x_0 is a **neighborhood** of x_0.

By the triangle inequality, an ϵ-neighborhood is open: if $d(x, x_0) < \epsilon$, then $U(x; \epsilon - d(x, x_0)) \subset U(x_0; \epsilon)$.

Theorem 3.1

Let (X, d) be a metric space. Then finite intersections and arbitrary unions of open sets of X are open in X. Similarly, finite unions and arbitrary intersections of closed sets of X are closed in X.

Proof [Supplement]. If x lies in a union of open sets, it lies in one of them, U_α, and some $U(x;\epsilon) \subset U_\alpha$ is contained in the union. If $x \in U_1 \cap \cdots \cap U_k$, choose ϵ_i with $U(x;\epsilon_i) \subset U_i$ and let $\epsilon = \min \epsilon_i > 0$. The statements about closed sets follow by De Morgan's laws. ■

Theorem 3.2

Let X be a metric space and Y a subspace. A subset A of Y is open in Y if and only if $A = U \cap Y$ for some U open in X. Similarly, A is closed in Y if and only if $A = C \cap Y$ for some C closed in X.

Proof [Supplement]. The ϵ-neighborhood of $y \in Y$ in Y is $U_X(y;\epsilon) \cap Y$. If U is open in X and $y \in U \cap Y$, then $U_X(y;\epsilon) \subset U$ for some ϵ, so $U_Y(y;\epsilon) \subset U \cap Y$. Conversely, if A is open in Y, choose for each $y \in A$ an ϵ_y with $U_X(y;\epsilon_y) \cap Y \subset A$ and put $U = \bigcup_{y\in A} U_X(y;\epsilon_y)$; then U is open in X and $U \cap Y = A$. For closed sets take complements: $Y - A = (X - C) \cap Y$. ■

Hence if A is open in Y and Y is open in X, then A is open in X; likewise for closed sets.

Definition 3.3 (limit point, closure)

$x_0 \in X$ is a **limit point** of $A \subset X$ if every ϵ-neighborhood of x_0 meets A in a point other than x_0 (equivalently, in infinitely many points of A). The **closure** $\bar{A}$ of A is A together with all its limit points.

Theorem 3.3

If A is a subset of X, then $\bar{A}$ is a closed set of X. A subset of X is closed if and only if it contains all its limit points.

Proof [Supplement]. A set C is closed iff every $x \notin C$ has a neighborhood disjoint from C, i.e. iff no point outside C is a limit point of C; this is the second statement. For the first: if $x \notin \bar{A}$, some $U(x;\epsilon)$ misses A. This open set also misses every limit point y of A: a point $y \in U(x;\epsilon)$ has a neighborhood inside $U(x;\epsilon)$, which contains no point of A. So $U(x;\epsilon) \cap \bar{A} = \emptyset$ and $X - \bar{A}$ is open. ■

In $\mathbb{R}^n$, the ϵ-neighborhood of $\mathbf{a}$ in the euclidean metric is the **open ball** $B(\mathbf{a};\epsilon)$; in the sup metric it is the **open cube** $C(\mathbf{a};\epsilon)$. By (1.1),

$$B(\mathbf{a};\epsilon) \subset C(\mathbf{a};\epsilon) \subset B(\mathbf{a};\sqrt{n}\,\epsilon).$$

Theorem 3.4

If X is a subspace of $\mathbb{R}^n$, the collection of open sets of X is the same whether one uses the euclidean metric or the sup metric on X. The same is true for the collection of closed sets of X.

Proof [Supplement]. If A is open for the sup metric and $\mathbf{x} \in A$, then $C(\mathbf{x};\epsilon) \cap X \subset A$ for some ϵ, hence $B(\mathbf{x};\epsilon) \cap X \subset A$. If A is open for the euclidean metric, $B(\mathbf{x};\epsilon) \cap X \subset A$, hence $C(\mathbf{x};\epsilon/\sqrt{n}) \cap X \subset A$. Closed sets are the complements. ■

A property of X depending only on its collection of open sets is a **topological property**; limits, continuity and compactness are examples.

3.2 Limits and continuity

Definition 3.4 (continuity)

Let X, Y be metric spaces. $f : X \to Y$ is **continuous at** x_0 if for each open $V \ni f(x_0)$ there is an open $U \ni x_0$ with $f(U) \subset V$; equivalently, for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), f(x_0)) < \epsilon$ whenever $d_X(x, x_0) < \delta$. f is **continuous** if it is continuous at every point; equivalently, $f^{-1}(V)$ is open in X for each open V of Y (or $f^{-1}(D)$ is closed for each closed D).

If some δ-neighborhood of x_0 is $\{x_0\}$, then x_0 is an **isolated point** of X, and every f is continuous at x_0. Constant maps and the identity are continuous.

Theorem 3.5

(a) Let $x_0 \in A$, where A is a subspace of X. If $f : X \to Y$ is continuous at x_0, then $f|A : A \to Y$ is continuous at x_0.
(b) Let $f : X \to Y$ and $g : Y \to Z$. If f is continuous at x_0 and g is continuous at $y_0 = f(x_0)$, then $g \circ f : X \to Z$ is continuous at x_0.

Proof [Supplement]. (a) The same δ works for points of A.

(b) Given ϵ, choose η with $d_Z(g(y), g(y_0)) < \epsilon$ for $d_Y(y, y_0) < \eta$, then δ with $d_Y(f(x), y_0) < \eta$ for $d_X(x, x_0) < \delta$. ■

Theorem 3.6

(a) Let X be a metric space and $f : X \to \mathbb{R}^n$, $f(x) = (f_1(x), \ldots, f_n(x))$. Then f is continuous at x_0 if and only if each **component function** $f_i : X \to \mathbb{R}$ is continuous at x_0.
(b) Let $f, g : X \to \mathbb{R}$ be continuous at x_0. Then $f + g$, $f - g$ and $f \cdot g$ are continuous at x_0, and f/g is continuous at x_0 if $g(x_0) \neq 0$.
(c) The projection $\pi_i : \mathbb{R}^n \to \mathbb{R}$, $\pi_i(\mathbf{x}) = x_i$, is continuous.

Proof [Supplement]. (a) Use the sup metric: $|f(x) - f(x_0)| < \epsilon$ iff $|f_i(x) - f_i(x_0)| < \epsilon$ for all i; take the minimum of the n values of δ.

(c) $|\pi_i(\mathbf{x}) - \pi_i(\mathbf{y})| \leq |\mathbf{x} - \mathbf{y}|$.

(b) Addition, subtraction and multiplication $\mathbb{R}^2 \to \mathbb{R}$ are continuous, and so is division on $\{(s, t) : t \neq 0\}$, by the usual estimates, e.g. $|st - s_0t_0| \leq |s||t - t_0| + |t_0||s - s_0|$. Compose with $x \mapsto (f(x), g(x))$, continuous by (a), and use Theorem 3.5. ■

Thus functions built from the familiar continuous functions by algebraic operations and composition are continuous, e.g. $f(s, t, u, v) = \sin(s + t)/e^{uv}$ on $\mathbb{R}^4$.

Definition 3.5 (limit)

Let $A \subset X$, $f : A \to Y$, and let x_0 be a limit point of A (not necessarily in A). We say $f(x) \to y_0$ as $x \to x_0$, or $\lim_{x \to x_0} f(x) = y_0$, if for each open $V \ni y_0$ there is an open $U \ni x_0$ with $f(x) \in V$ for all $x \in U \cap A$, $x \neq x_0$. Equivalently: for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), y_0) < \epsilon$ whenever $x \in A$ and $0 < d_X(x, x_0) < \delta$.

We do not attempt to define the limit if x_0 is not a limit point of the domain; the value of f at x_0 plays no role.

Theorem 3.7

Let $f : X \to Y$. If x_0 is an isolated point of X, then f is continuous at x_0. Otherwise, f is continuous at x_0 if and only if $f(x) \to f(x_0)$ as $x \to x_0$.

Proof [Supplement]. Compare the ϵ-δ conditions: they differ only in whether $x = x_0$ is allowed, and at $x = x_0$ the condition $d_Y(f(x), f(x_0)) < \epsilon$ holds trivially. ■

Theorem 3.8

(a) Let $A \subset X$ and $f : A \to \mathbb{R}^n$, $f = (f_1, \ldots, f_n)$; let $\mathbf{a} = (a_1, \ldots, a_n)$. Then $f(x) \to \mathbf{a}$ as $x \to x_0$ if and only if $f_i(x) \to a_i$ for each i.
(b) Let $f, g : A \to \mathbb{R}$. If $f(x) \to a$ and $g(x) \to b$ as $x \to x_0$, then $f+g \to a+b$, $f-g \to a-b$, $f \cdot g \to a \cdot b$, and $f/g \to a/b$ if $b \neq 0$.

Proof [Supplement]. Extend f, g to $A \cup \{x_0\}$ by $f(x_0) = \mathbf{a}$ (resp. a, b). The limit statements become continuity at x_0 of the extended functions (Theorem 3.7), and Theorem 3.6 applies. ■

3.3 Interior and exterior

Definition 3.6 (interior, exterior, boundary)

For $A \subset \mathbb{R}^n$: $\operatorname{Int} A$ is the union of all open sets of $\mathbb{R}^n$ contained in A; $\operatorname{Ext} A$ is the union of all open sets disjoint from A; $\operatorname{Bd} A$ consists of the points in neither $\operatorname{Int} A$ nor $\operatorname{Ext} A$.

A point $\mathbf{x}$ is in $\operatorname{Bd} A$ iff every open set containing $\mathbf{x}$ meets both A and $\mathbb{R}^n - A$. $\mathbb{R}^n$ is the disjoint union of $\operatorname{Int} A$, $\operatorname{Ext} A$ and $\operatorname{Bd} A$; the first two are open and $\operatorname{Bd} A$ is closed.

For a **rectangle** $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ one checks $\operatorname{Int} Q = (a_1, b_1) \times \cdots \times (a_n, b_n)$ (an **open rectangle**), $\operatorname{Ext} Q = \mathbb{R}^n - Q$, $\operatorname{Bd} Q = Q - \operatorname{Int} Q$. The open cube $C(\mathbf{a}; \epsilon) = (a_1 - \epsilon, a_1 + \epsilon) \times \cdots \times (a_n - \epsilon, a_n + \epsilon)$ is an open rectangle; the corresponding closed rectangle $[a_1 - \epsilon, a_1 + \epsilon] \times \cdots \times [a_n - \epsilon, a_n + \epsilon]$ is the **closed cube** (or simply cube) centered at $\mathbf{a}$.

§4. Compact Subspaces and Connected Subspaces of $\mathbb{R}^n$

Motivation. Compactness is what turns pointwise statements into uniform ones (uniform continuity, existence of maxima, Lebesgue numbers). These facts are used repeatedly in the existence of the integral and in the proof of the change of variables theorem.

4.1 Compact spaces

Definition 4.1 (compact)

Let $X \subset \mathbb{R}^n$. A **covering** of X is a collection of subsets of $\mathbb{R}^n$ whose union contains X; it is an **open covering** if each member is open in $\mathbb{R}^n$. X is **compact** if every open covering of X contains a finite subcollection that also covers X.

Theorem 4.1

A subspace X of $\mathbb{R}^n$ is compact if and only if for every collection of sets open in X whose union is X, there is a finite subcollection whose union equals X.

Proof. Suppose X is compact and $\{A_\alpha\}$ are open in X with union X. By Theorem 3.2 choose U_α open in $\mathbb{R}^n$ with $A_\alpha = U_\alpha \cap X$. The U_α cover X, so finitely many, $U_{\alpha_1}, \ldots, U_{\alpha_k}$, do; then $A_{\alpha_1}, \ldots, A_{\alpha_k}$ have union X. Conversely, given an open covering $\{U_\alpha\}$ of X by open sets of $\mathbb{R}^n$, the sets $U_\alpha \cap X$ are open in X with union X; a finite subfamily with union X gives finitely many U_α covering X. ■

Theorem 4.2

The subspace $[a, b]$ of $\mathbb{R}$ is compact.

Proof [Supplement]. Let $\mathcal{A}$ be an open covering of $[a, b]$ and let S be the set of $x \in [a, b]$ such that $[a, x]$ is covered by finitely many members of $\mathcal{A}$. Then $a \in S$; let $c = \sup S$. Some $U \in \mathcal{A}$ contains c, and with it an interval $(c - \delta, c + \delta)$. There is $x \in S$ with $x > c - \delta$; adding U to the finite family covering $[a, x]$ covers $[a, \min(c + \delta/2, b)]$. Hence $c \in S$, and if $c < b$ we would get a point of S larger than c. So $c = b \in S$. ■

Definition 4.2 (bounded)

$X \subset \mathbb{R}^n$ is **bounded** if there is M with $|\mathbf{x}| \leq M$ for all $\mathbf{x} \in X$.

Theorem 4.3

If X is a compact subspace of $\mathbb{R}^n$, then X is closed and bounded.

Proof.

Step 1. Bounded: the open cubes $U_N = C(\mathbf{0}; N)$, $N = 1, 2, \ldots$, are open, increasing, and cover $\mathbb{R}^n$, hence X. Finitely many, $U_{N_1}, \ldots, U_{N_k}$, cover X; with $M = \max N_i$, $X \subset U_M$.

Step 2. Closed: let $\mathbf{a} \notin X$; we find a neighborhood of $\mathbf{a}$ disjoint from X. Let $C_N = \{\mathbf{x} : |\mathbf{x} - \mathbf{a}| \leq 1/N\}$ and $V_N = \mathbb{R}^n - C_N$. The V_N are open, increasing, and cover $\mathbb{R}^n - \{\mathbf{a}\} \supset X$. Finitely many cover X, so $X \subset V_M$ for some M; then C_M is disjoint from X, and in particular the open cube $C(\mathbf{a}; 1/M)$ lies in $\mathbb{R}^n - X$. ■

Corollary 4.4

Let X be a compact subspace of $\mathbb{R}$. Then X has a largest element and a smallest element.

Proof. X is bounded, so it has a supremum and an infimum; X is closed, so these (being limit points of X or points of X) belong to X. ■

Theorem 4.5 (Extreme-value theorem)

Let X be a compact subspace of $\mathbb{R}^m$. If $f : X \to \mathbb{R}^n$ is continuous, then $f(X)$ is a compact subspace of $\mathbb{R}^n$. In particular, if $\phi : X \to \mathbb{R}$ is continuous, then ϕ has a maximum value and a minimum value.

Proof. Let $\{V_\alpha\}$ be an open covering of $f(X)$. The sets $f^{-1}(V_\alpha)$ are open in X and cover X, so by Theorem 4.1 finitely many, for $\alpha = \alpha_1, \ldots, \alpha_k$, cover X; then $V_{\alpha_1}, \ldots, V_{\alpha_k}$ cover $f(X)$. If $\phi : X \to \mathbb{R}$ is continuous, $\phi(X)$ is compact, and by Corollary 4.4 it has a largest and a smallest element. ■

Definition 4.3 (ϵ-neighborhood of a set)

For $X \subset \mathbb{R}^n$ and $\epsilon > 0$, the union of the balls $B(\mathbf{a};\epsilon)$, $\mathbf{a} \in X$, is the **ϵ-neighborhood** of X in the euclidean metric; the union of the cubes $C(\mathbf{a};\epsilon)$ is the ϵ-neighborhood in the sup metric.

Theorem 4.6 (The ϵ-neighborhood theorem)

Let X be a compact subspace of $\mathbb{R}^n$; let U be an open set of $\mathbb{R}^n$ containing X. Then there is an $\epsilon > 0$ such that the ϵ-neighborhood of X (in either metric) is contained in U.

Proof. The euclidean ϵ-neighborhood lies inside the sup-metric one, so it suffices to treat the sup metric.

Step 1. For $C \subset \mathbb{R}^n$ define $d(\mathbf{x}, C) = \inf\{|\mathbf{x} - \mathbf{c}| : \mathbf{c} \in C\}$, the **distance from x to** C. It is continuous: for $\mathbf{c} \in C$, $d(\mathbf{x}, C) - |\mathbf{x} - \mathbf{y}| \le |\mathbf{x} - \mathbf{c}| - |\mathbf{x} - \mathbf{y}| \le |\mathbf{y} - \mathbf{c}|$; taking the infimum over $\mathbf{c}$, $d(\mathbf{x}, C) - d(\mathbf{y}, C) \le |\mathbf{x} - \mathbf{y}|$, and symmetrically.

Step 2. Let $f(\mathbf{x}) = d(\mathbf{x}, \mathbb{R}^n - U)$ for $\mathbf{x} \in X$. Then f is continuous and positive on X: if $\mathbf{x} \in X$, some $C(\mathbf{x};\delta) \subset U$, so $f(\mathbf{x}) \ge \delta$. Since X is compact, f has a minimum value $\epsilon > 0$ (Theorem 4.5). Then for $\mathbf{a} \in X$, every point $\mathbf{y}$ with $|\mathbf{y} - \mathbf{a}| < \epsilon$ lies in U; i.e. the ϵ-neighborhood of X lies in U. ■

Some hypothesis on X is needed: if X is the x-axis in $\mathbb{R}^2$ and $U = \{(x, y) : y^2 < 1/(1 + x^2)\}$, no ϵ-neighborhood of X lies in U.

Theorem 4.7 (Uniform continuity)

Let X be a compact subspace of $\mathbb{R}^m$; let $f : X \to \mathbb{R}^n$ be continuous. Given $\epsilon > 0$, there is a $\delta > 0$ such that whenever $\mathbf{x}, \mathbf{y} \in X$,

$$|\mathbf{x} - \mathbf{y}| < \delta \quad \text{implies} \quad |f(\mathbf{x}) - f(\mathbf{y})| < \epsilon.$$

This result also holds if one uses the euclidean metric instead of the sup metric.

The conclusion is the condition of **uniform continuity**.

Proof. Consider $X \times X \subset \mathbb{R}^m \times \mathbb{R}^m$ and the **diagonal** $\Delta = \{(\mathbf{x}, \mathbf{x}) : \mathbf{x} \in X\}$, which is compact as the image of X under the continuous map $\mathbf{x} \mapsto (\mathbf{x}, \mathbf{x})$. First use the euclidean metric. The function $g : X \times X \to \mathbb{R}$, $g(\mathbf{x}, \mathbf{y}) = \|f(\mathbf{x}) - f(\mathbf{y})\|$, is continuous, so $\{(\mathbf{x}, \mathbf{y}) : g(\mathbf{x}, \mathbf{y}) < \epsilon\}$ is open in $X \times X$ and contains Δ; hence it equals $U \cap (X \times X)$ for some open $U \subset \mathbb{R}^{2m}$ containing Δ (Theorem 3.2). By Theorem 4.6, some δ-neighborhood of Δ lies in U. If $\mathbf{x}, \mathbf{y} \in X$ with $\|\mathbf{x} - \mathbf{y}\| < \delta$, then $\|(\mathbf{x}, \mathbf{y}) - (\mathbf{y}, \mathbf{y})\| = \|(\mathbf{x} - \mathbf{y}, \mathbf{0})\| = \|\mathbf{x} - \mathbf{y}\| < \delta$, so $(\mathbf{x}, \mathbf{y})$ lies in the δ-neighborhood of Δ, hence in U, and $g(\mathbf{x}, \mathbf{y}) < \epsilon$.

For the sup metric: if $|\mathbf{x} - \mathbf{y}| < \delta/\sqrt{m}$ then $\|\mathbf{x} - \mathbf{y}\| < \delta$, whence $|f(\mathbf{x}) - f(\mathbf{y})| \le \|f(\mathbf{x}) - f(\mathbf{y})\| < \epsilon$. ■

Lemma 4.8

The rectangle $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ in $\mathbb{R}^n$ is compact.

Proof. Induction on n; the case $n = 1$ is Theorem 4.2. Write $Q = X \times [a_n, b_n]$ with X a rectangle in $\mathbb{R}^{n-1}$, compact by the induction hypothesis. Let $\mathcal{A}$ be an open covering of Q.

Step 1. Given $t \in [a_n, b_n]$, there is $\epsilon > 0$ such that $X \times (t - \epsilon, t + \epsilon)$ is covered by finitely many elements of $\mathcal{A}$. Indeed, $X \times t$ is compact (the image of X under $\mathbf{x} \mapsto (\mathbf{x}, t)$), so it is covered by finitely many $A_1, \ldots, A_k \in \mathcal{A}$. Their union U is open and contains $X \times t$; by Theorem 4.6 some ϵ-neighborhood of $X \times t$ lies in U, and in particular $X \times (t - \epsilon, t + \epsilon) \subset U$.

Step 2. By Step 1 choose for each t an open interval $V_t \ni t$ such that $X \times V_t$ is covered by finitely many elements of $\mathcal{A}$. The V_t cover $[a_n, b_n]$, so finitely many, $V_{t_1}, \ldots, V_{t_m}$, do. Then Q is contained in $\bigcup_i X \times V_{t_i}$, each covered by finitely many elements of $\mathcal{A}$; so Q is covered by finitely many. ■

Theorem 4.9

If X is a closed and bounded subspace of $\mathbb{R}^n$, then X is compact.

Proof. Let $\mathcal{A}$ be an open covering of X. Adjoin the open set $\mathbb{R}^n - X$ to get an open covering of $\mathbb{R}^n$. Since X is bounded, it lies in a rectangle Q, which is compact (Lemma 4.8); finitely many members of the enlarged collection cover Q. Discarding $\mathbb{R}^n - X$ if it occurs leaves finitely many members of $\mathcal{A}$; they may not cover Q, but they cover X, since $\mathbb{R}^n - X$ contains no point of X. ■

All theorems of this section hold for arbitrary metric spaces in place of $\mathbb{R}^n, \mathbb{R}^m$, *except* Theorem 4.9.

4.2 Connected spaces

Definition 4.4 (connected)

A metric space X is **connected** if X cannot be written as the union of two disjoint non-empty sets A and B, each of which is open in X.

Theorem 4.10

The closed interval $[a, b]$ of $\mathbb{R}$ is connected.

Proof [Supplement]. Suppose $[a, b] = A \cup B$ with A, B disjoint, non-empty and open in $[a, b]$ (hence also closed in $[a, b]$); say $a \in A$. Let $c = \sup\{x \in [a, b] : [a, x] \subset A\}$. Since A is closed, $c \in A$. If $c < b$, openness of A gives $[c, c + \delta) \subset A$ for some $\delta > 0$, so $[a, c + \delta/2] \subset A$, contradicting the definition of c. So $c = b$ and $[a, b] \subset A$, so $B = \emptyset$, a contradiction. ■

Theorem 4.11 (Intermediate-value theorem)

Let X be connected. If $f : X \to Y$ is continuous, then $f(X)$ is a connected subspace of Y. In particular, if $\phi : X \to \mathbb{R}$ is continuous and $f(x_0) < r < f(x_1)$ for some points x_0, x_1 of X, then $f(x) = r$ for some point x of X.

Proof. If $f(X) = A \cup B$ with A, B disjoint and open in $f(X)$, then $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint, open in X, with union X; connectedness of X forces one of them, hence one of A, B, to be empty. For ϕ: let $A = \{y \in \mathbb{R} : y < r\}$, $B = \{y : y > r\}$, open in $\mathbb{R}$. If $r \notin \phi(X)$, then $\phi(X)$ is the union of the disjoint sets $\phi(X) \cap A$, $\phi(X) \cap B$, open in $\phi(X)$ and both non-empty, contradicting connectedness of $\phi(X)$. ■

Definition 4.5 (line segment, convex)

The **line segment** joining $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is the set of points $\mathbf{x} = \mathbf{a} + t(\mathbf{b} - \mathbf{a})$, $0 \le t \le 1$. A set $A \subset \mathbb{R}^n$ is **convex** if for every pair $\mathbf{a}, \mathbf{b} \in A$ the segment joining them lies in A.

A line segment is connected, being the image of $[0, 1]$ under $t \mapsto \mathbf{a} + t(\mathbf{b} - \mathbf{a})$. Every convex set A is connected: if $A = U \cup V$ with U, V disjoint, non-empty and open in A, pick $\mathbf{a} \in U$, $\mathbf{b} \in V$ and let L be the segment joining them; then $U \cap L$ and $V \cap L$ are disjoint, non-empty, open in L, with union L, a contradiction. Hence open balls, open cubes and rectangles in $\mathbb{R}^n$ are connected.